

Optimising in vitro techniques to study astrocyte on a molecular and cellular level

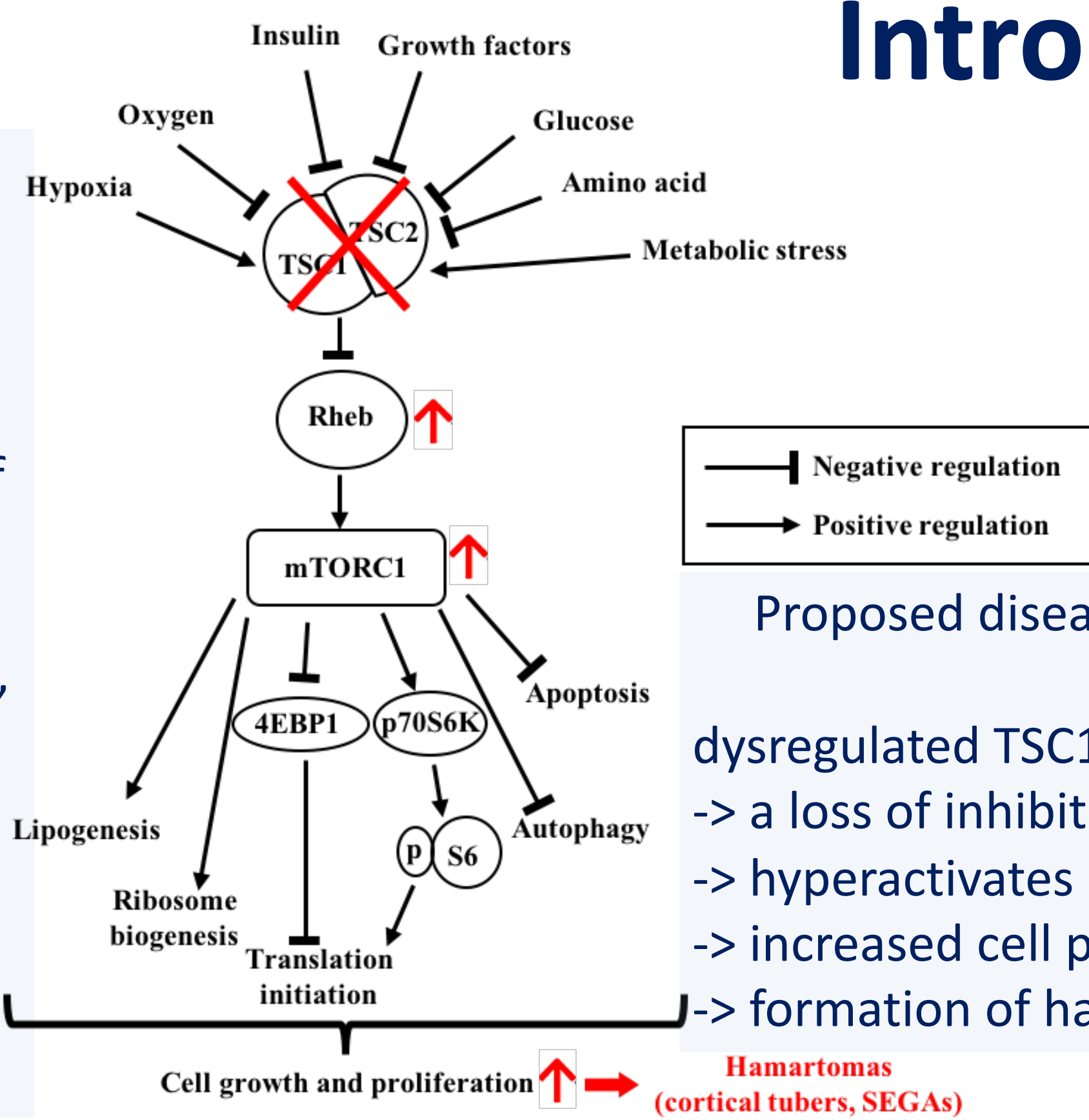
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Introduction

Tuberous sclerosis complex

- A neurological disorder characterised by the formation of hamartomas
- Caused by mutations in *TSC1/2*
- Show impaired glutamate uptake, commonly resulting in epilepsy (60 - 90%) [1]
- Show evidence for astrocyte dysfunction arising from hyperactivation of the mTOR pathway



Overall aim: Use rat astrocytes to optimise assays for developing an *in vitro* system to examine the mTOR pathway and glial functioning in astrocytes in the context of TSC

Molecular

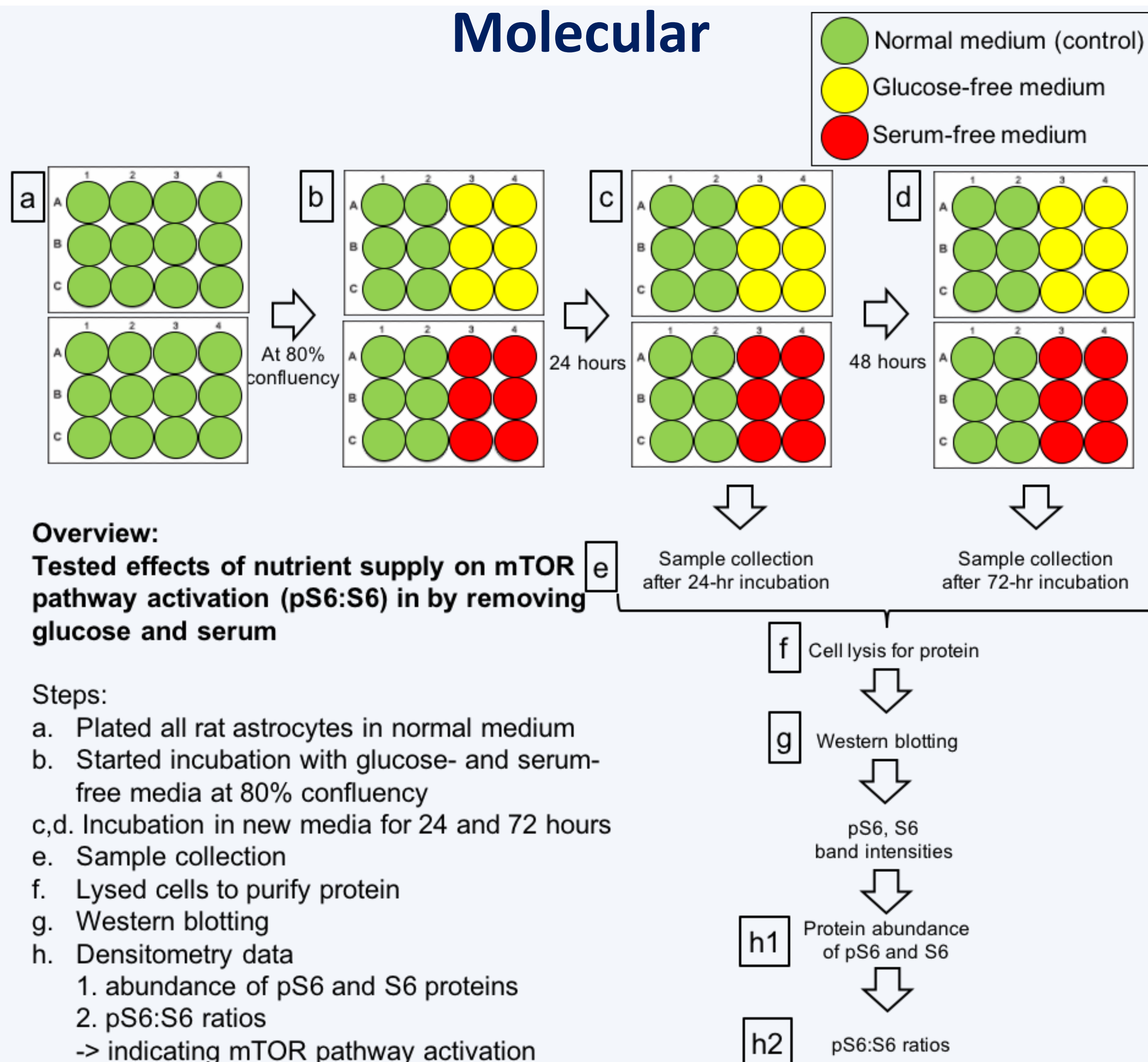
- High glucose concentrations mask cellular TSC phenotypes in an iPSC model [2]
- It is unknown whether this glucose-interfering pattern also appears in astrocytes in TSC
- Aim (i): assess effects of nutrient supply on mTORC1 activation in rat astrocytes for assay optimization

Cellular

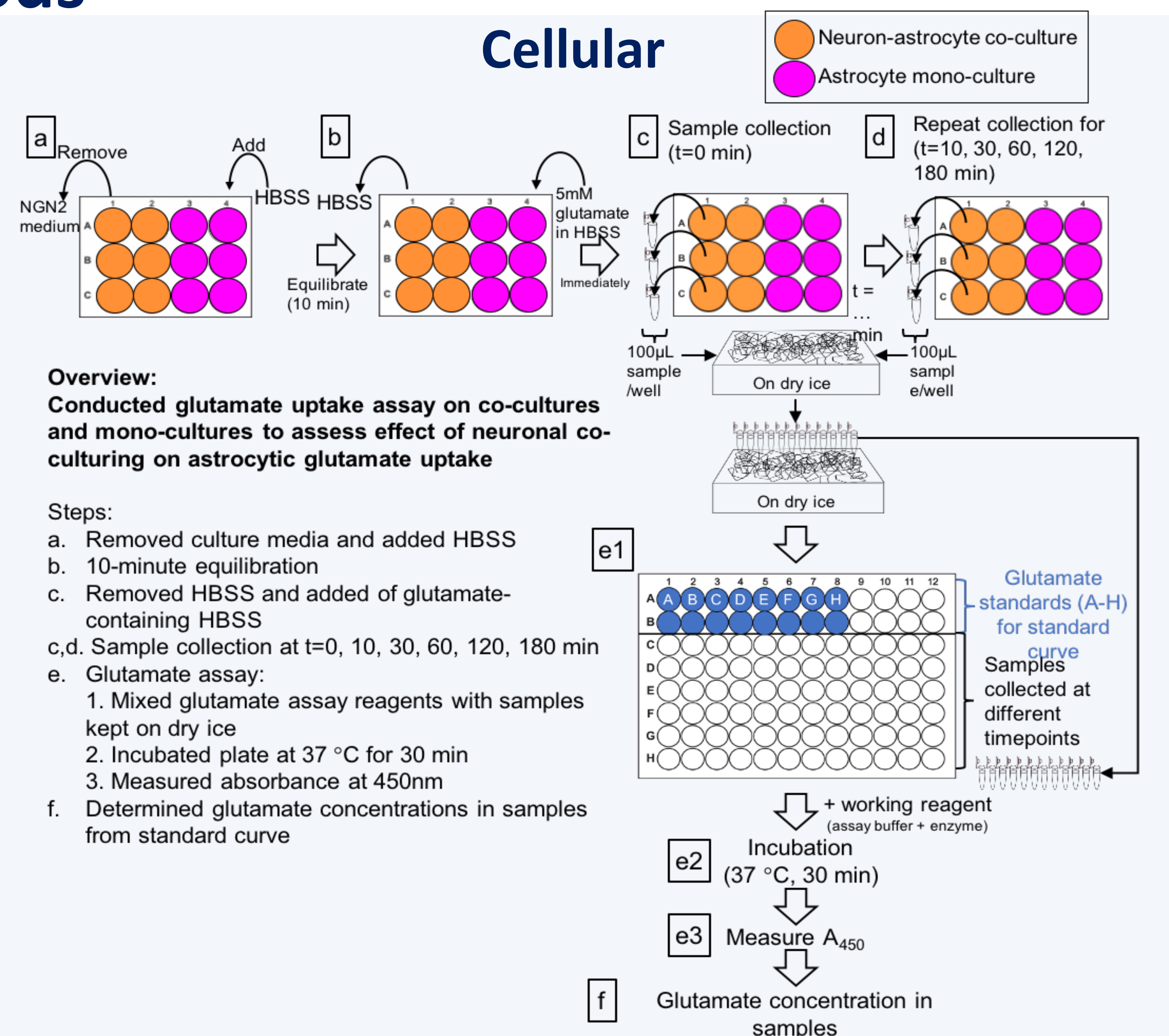
- enhancing effects of glial co-culture to neuronal function and survival is well-established [3]
- it is unknown astrocytes benefit from neuronal presence
- Previous studies have shown neuronal presence is required astrocytic glutamate transporter expression to protect neurons from glutamate toxicity [4,5]
- Aim (ii): evaluate effect of neuronal presence on glial functioning to optimize glutamate uptake assay

Methods

Molecular

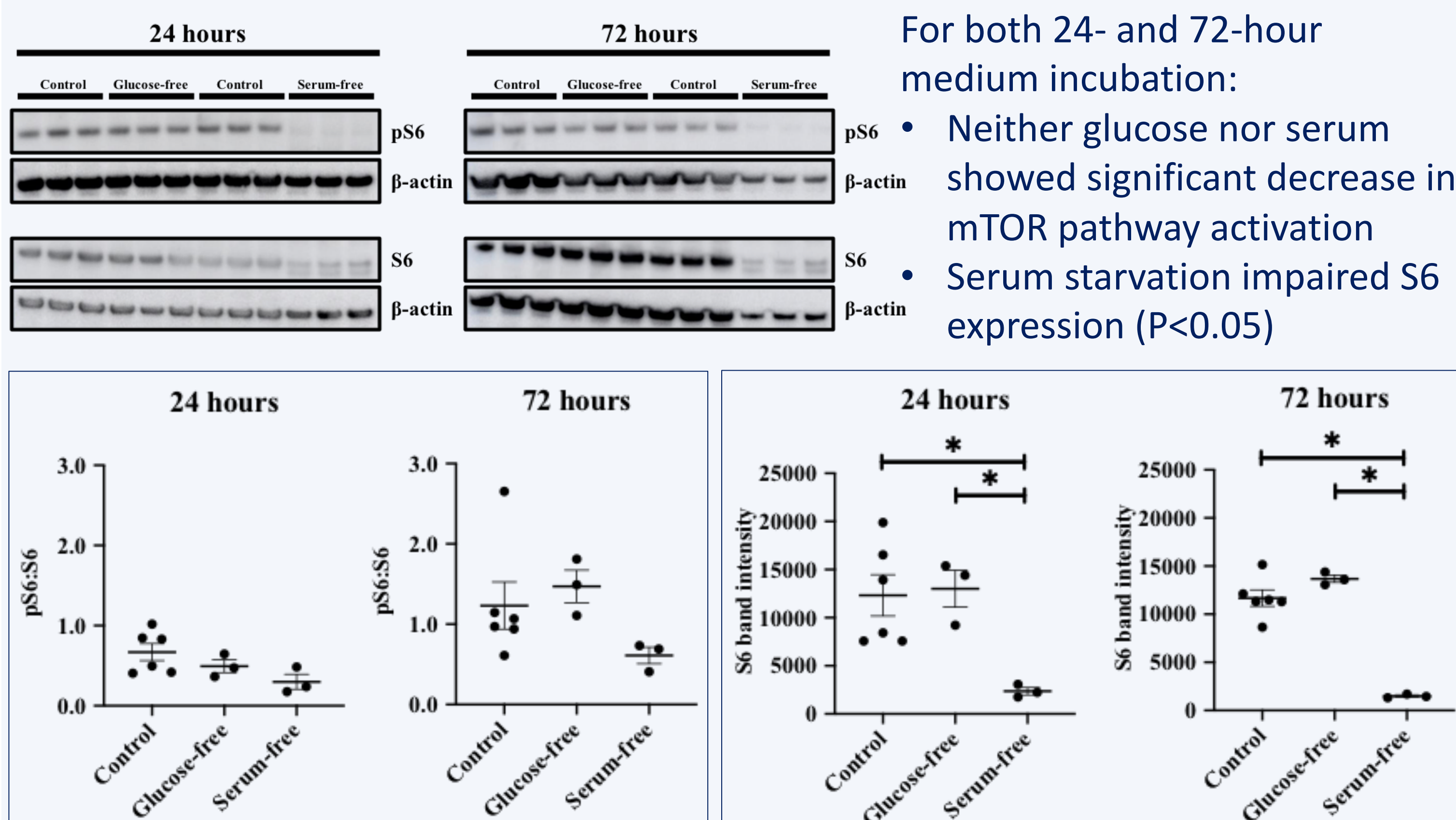


Cellular

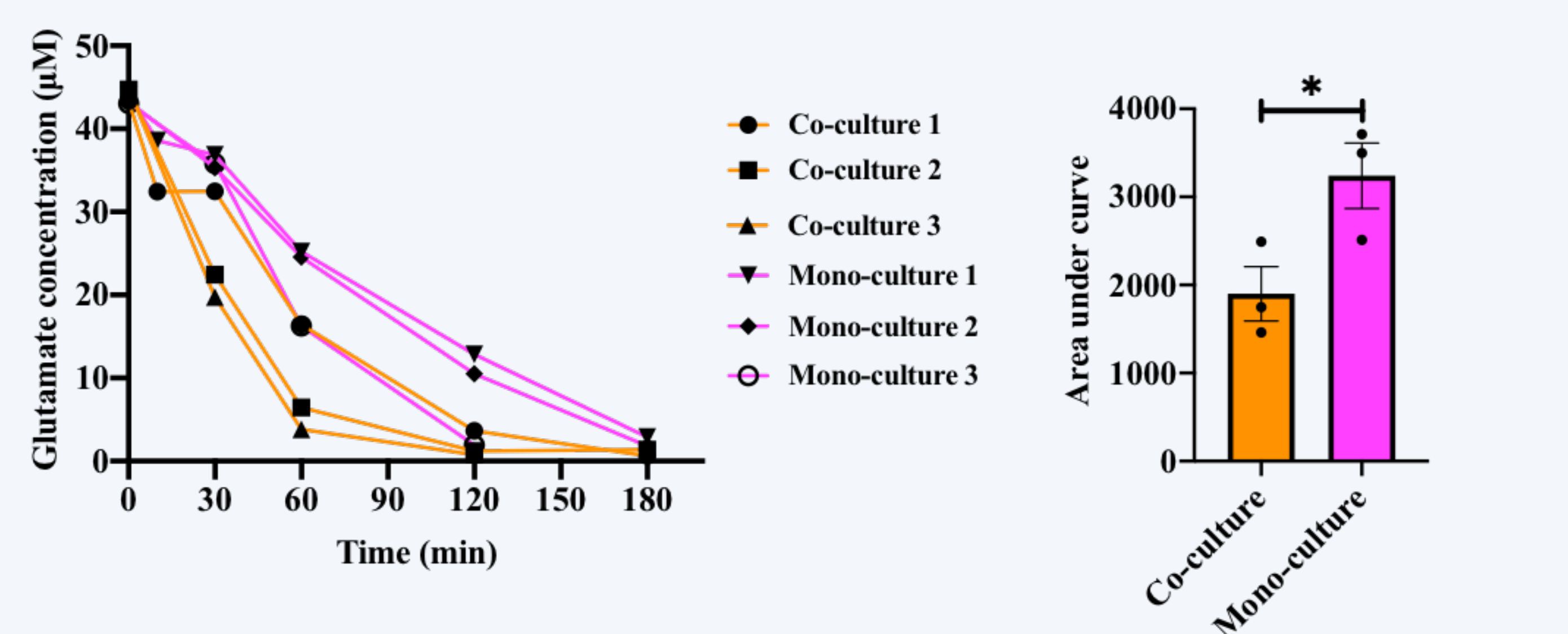


Results

Molecular



Cellular



- Co-cultures showed a higher rate of decline in extracellular glutamate concentrations than mono-cultures
- Mean AUC for co-culture glutamate uptake was significantly lower than mono-culture uptake (P<0.05)
- Neurone-astrocyte co-cultures showed a higher rate of glutamate uptake than mono-cultures

Conclusions

Molecular

- Neuronal co-culture promoted glutamate uptake function of astrocytes
- Future experiments could measure expression of glutamate transporters to confirm glutamate uptake is predominantly mediated by astrocytes

Cellular

- Serum seems to be required for normal S6 expression
- Further work required to confirm the lack of change in mTOR pathway activation in response to loss of nutrients, and test whether the lack of S6 expression was mTOR-dependent (e.g. by applying an mTOR activator, MHY1485)

Overall, astrocytic research remains a novel field requiring standardization in methods to generate astrocytes and optimised assays for research. More endeavours in researching on mTOR pathway and astrocytes will hopefully increase the understanding of their roles and uncover potential therapeutic targets for diseases such as TSC.

References

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- [3] Brown, D.R., 1999. Neurons depend on astrocytes in a coculture system for protection from glutamate toxicity. *Molecular and cellular neuroscience*, 13(5), pp.379–389.
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- [5] Swanson, R. A., Liu, J., Miller, J. W., Rothstein, J. D., Farrell, K., Stein, B. A., & Longuemare, M. C. (1997). Neuronal regulation of glutamate transporter subtype expression in astrocytes. *Journal of Neuroscience*, 17(3), 932–940.

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